

Statistical Field Theory: Example Sheet 1

Harvey Reall, Michaelmas 2025

Comments and corrections to hsr1000@cam.ac.uk.

1. The purpose of this question is to solve the 1d Ising model with N spins, using periodic boundary conditions such that $s_{N+1} = s_1$. The partition function is given by

$$Z = \sum_{s_1=\pm 1} \dots \sum_{s_N=\pm 1} \prod_i \exp \left(\beta J s_i s_{i+1} + \frac{\beta B}{2} (s_i + s_{i+1}) \right)$$

Show that the partition can be written in terms of the 2×2 transfer matrix T as

$$Z = \text{tr } T^N \quad \text{with} \quad T = \begin{pmatrix} e^{\beta J - \beta B} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J + \beta B} \end{pmatrix}$$

[Hint: Think of the i^{th} transfer matrix in the string T^N using index notation, written as $T_{s_i, s_{i+1}} = \exp \left(\beta J s_i s_{i+1} + \frac{\beta B}{2} (s_i + s_{i+1}) \right)$. Here $s_i = \pm 1$ labels the column and $s_{i+1} = \pm 1$ labels the row.]

Show that the eigenvalues of T are given by

$$\lambda_{\pm} = e^{\beta J} \cosh \beta B \pm \sqrt{e^{2\beta J} \cosh^2 \beta B - 2 \sinh 2\beta J}$$

and hence that, as $N \rightarrow \infty$, the partition function is $Z \approx \lambda_+^N$. Use this to show that, for $B = 0$, the magnetisation always vanishes and hence that there is no phase transition as a function of temperature.

2. This question provides an alternative, mean field approach to the Ising model. (This is, in fact, the original approach, first suggested by Weiss.) Write the interaction terms in the Ising model as

$$s_i s_j = (s_i - m)(s_j - m) + m(s_j - m) + m(s_i - m) + m^2$$

where m is the equilibrium magnetisation. The mean field “approximation” hinges on assuming that the deviation from equilibrium, $(s_i - m)$, is small and so the first term above can be neglected. With this assumption, show that the partition function of the Ising model can be written as

$$Z = \sum_{\{s_i\}} e^{-\frac{1}{2}\beta J N q m^2 + \beta(J q m + B) \sum_i s_i} = e^{-\frac{1}{2}\beta J N q m^2} 2^N \cosh^N(\beta J q m + \beta B)$$

with q the number of nearest neighbour pairs. Use this partition function to determine the equilibrium magnetisation and show that it obeys the self-consistency condition

$$m = \tanh(\beta B + \beta J q m)$$

For $B = 0$, show that the form of the solution differs for $T > T_c$ and $T < T_c$ where $T_c = Jq$. [*Hint*: Think graphically.] For $B \neq 0$, find the asymptotic form of the solution as $T \rightarrow \infty$.

3. This question describes a more rigorous approach to mean field theory. First recall that the Gibbs entropy of a probability distribution $\{p_i\}$ is defined as $S = -\sum_i p_i \log p_i$. For a physical system with energy levels E_i define the “variational free energy” of the probability distribution as $\mathcal{F} = \langle E \rangle - TS$ where $\langle E \rangle = \sum_i E_i p_i$ is the expectation value of the energy and T is the temperature. Show that minimizing $\mathcal{F}$ w.r.t. the probability distribution leads to the Boltzmann distribution. Hence the minimum of $\mathcal{F}$ over all probability distributions is the usual free energy F .

This motivates a variational approximation for statistical physics problems: start from an Ansatz for the probability distribution $\{p_i\}$ containing some free parameters and then minimize $\mathcal{F}$ w.r.t. these parameters. Let’s apply this to the Ising model. Mean field theory corresponds to the Ansatz of a “factorized” probability distribution, where each spin behaves independently of its neighbours with $p(s_i = \pm 1) = (1 \pm m)/2$ where $m \in [-1, 1]$ is a parameter. Show that $\langle s_i \rangle = m$ and $\langle s_i s_j \rangle = m^2$ and hence show that $\mathcal{F}(m)$ is the same as the effective free energy of mean field theory. Finally show that minimizing w.r.t. m reproduces the self-consistency condition of the previous question.

4. The long range Ising model, in which each spin interacts with all other spins, including itself, provides an example where mean field theory is exact. The energy is

$$E = -B \sum_i s_i - \frac{J}{2N} \left(\sum_i s_i \right) \left(\sum_j s_j \right)$$

Why is there an extra factor of N in the coupling? First, show that

$$e^{\beta J \alpha^2 / 2N} = \sqrt{\frac{N\beta J}{2\pi}} \int_{-\infty}^{+\infty} dx e^{-N\beta J x^2 / 2 + \alpha\beta J x}$$

Use this identity to write the partition function as

$$Z = \sqrt{\frac{N\beta J}{2\pi}} \int_{-\infty}^{+\infty} dx e^{-NS(x)} \quad \text{where} \quad S(x) = \frac{\beta J x^2}{2} - \log 2 \cosh \beta(B + Jx)$$

In the large N limit, the partition function can be evaluated by saddle point so that, ignoring a pre-factor, $Z \approx e^{-NS(x_*)}$ where x_* is the minimum of $S(x)$. Find an equation for x_* . Show that the equilibrium magnetisation is given by $m = x_*$.

5. (*) Consider the free energy

$$f(m) = \alpha_2(T)m^2 + \alpha_4 m^4 + \alpha_6 m^6$$

where $\alpha_4 < 0$ and $\alpha_6 > 0$ and $\alpha_2(T)$ varies from positive to negative as the temperature is lowered. Sketch $f(m)$ for various values of the temperature. Show that the system undergoes a first order phase transition at $\alpha_2 = \alpha_4^2/4\alpha_6$. What is the jump in the magnetisation? Draw the phase diagram in the $\alpha_2 - \alpha_4$ plane.

When $\alpha_4 = 0$, the phase transition occurs at $\alpha_2(T_c) = 0$. This is said to be a *tri-critical point*. Compute the mean field critical exponents α , β , γ and δ . [You may assume that $\alpha_2(T) \sim (T - T_c)$ close to T_c and should recall that for two of the critical exponents a $-Bm$ term needs to be added to $f(m)$.]

6. A superfluid has complex order parameter ψ and free energy given by the so-called XY-model

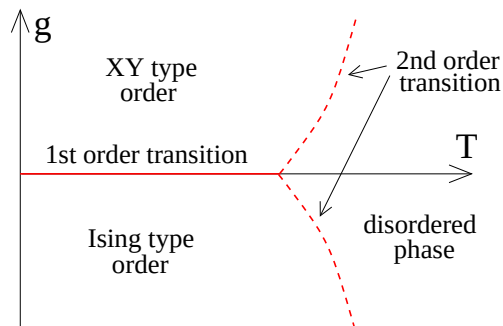
$$f(\psi) = \alpha_2 |\psi|^2 + \alpha_4 |\psi|^4$$

Sketch the free energy for $\alpha_2 < 0$ and $\alpha_4 > 0$. What are the possible ground states when $\alpha_2 < 0$? What is the meaning of “spontaneous symmetry breaking” in this context? Show that the critical point at $\alpha_2 = 0$ has the same mean field exponents as the Ising model. (To define some of these you need to couple to a complex B -field by adding the term $-\text{Re}(\bar{B}\psi)$ to f .)

7. Consider a lattice model where there lives, on each site, a unit 3-vector $\mathbf{s} = (s^x, s^y, s^z)$ with $|\mathbf{s}| = 1$. The energy is given by

$$E = -J \sum_{\langle ij \rangle} \mathbf{s}_i \cdot \mathbf{s}_j + g \sum_i \left((s_i^z)^2 - \frac{1}{2} \left((s_i^x)^2 + (s_i^y)^2 \right) \right)$$

with $J > 0$. Use mean field theory to give plausibility arguments for why the phase diagram takes the form shown below.



[*Hint:* Consider the ground states, and the phase transitions, between them in the following regimes: (i) At low T , as g changes from negative to positive. (ii) With $g > 0$ as T changes from low to high. (iii) With $g < 0$ as T changes from low to high.]

8. (*) Consider the Landau-Ginsberg free energy for a complex scalar ψ ,

$$F = \int dx \left(\alpha_2(T) |\psi|^2 + \alpha_4 |\psi|^4 - \gamma \left| \frac{d\psi}{dx} \right|^2 + \kappa \left| \frac{d^2\psi}{dx^2} \right|^2 \right)$$

with $\alpha_4, \gamma, \kappa > 0$. Consider an ansatz in which a single Fourier mode $\psi_k = A_k e^{ikx}$ with $k = \pm k_0$ is non-vanishing. What value of k_0 minimises the free energy? Show that the system undergoes a phase transition to a spatially modulated phase when $\alpha_2 = \gamma^2/4\kappa$. What symmetries are broken in the ordered phase?

9. A multi-critical point arises from the free energy

$$f(m) = \alpha_2 m^2 + \alpha_{2n} m^{2n}, \quad n \in \mathbf{Z}^+$$

Determine the mean field critical exponent $m \sim (T_c - T)^\beta$ in the ordered phase. Substitute this back into the free energy to show that the mean field critical exponent $c \sim (T_c - T)^{-\alpha}$ in the ordered phase is given by $\alpha = 1 - 2\beta$. Compare this with the contribution to the free energy from fluctuations in the Gaussian path integral. Show that the mean field contribution to the heat capacity dominates at the critical point provided $d > d_c = 2n/(n - 1)$.